📊 Pricing Strategy Analysis – Complete Workflow

🔹 1. Problem Definition

The first step was to clearly define the business objective:

- What is the optimal pricing strategy?
- How does price affect demand (elastic vs inelastic)?
- What pricing actions can maximize revenue and profit?

Goal:

- Maximize revenue
- Optimize profit margins

🔹 2. Data Collection

A structured dataset was used containing:

- Product Name
- Unit Price
- Quantity Sold
- Revenue
- Cost Price
- Competitor Price
- Price Elasticity

This ensures a **data-driven pricing decision approach**, similar to standard analytics workflows

---

🔹 3. Data Cleaning (Excel)

Performed preprocessing in Excel:

- Removed missing/null values
- Standardized price formats
- Eliminated duplicates
- Cleaned inconsistent product/category names

Outcome: Clean and reliable dataset for analysis

◈ 4. Feature Engineering

Created key business metrics:

- Revenue = Price × Quantity
- Profit = Revenue – Cost
- Profit Margin (%)
- Price Elasticity

This step transformed raw data into **decision-making metrics**.

◈ 5. Exploratory Data Analysis (Python)

Used Python (Pandas, Matplotlib) to analyze:

- Price vs Quantity (Demand behavior)
- Revenue trends
- Elastic vs Inelastic products
- Competitor price comparison

Insights Identified:

- Which products are price-sensitive
- Which products support premium pricing

◈ 6. Pricing Strategy Design

Based on analysis, the following strategies were developed:

◆ **Discount Strategy**

- Apply discounts to elastic products
- Example: Wireless Headphones → 10–15% discount

◆ **Bundle Strategy**

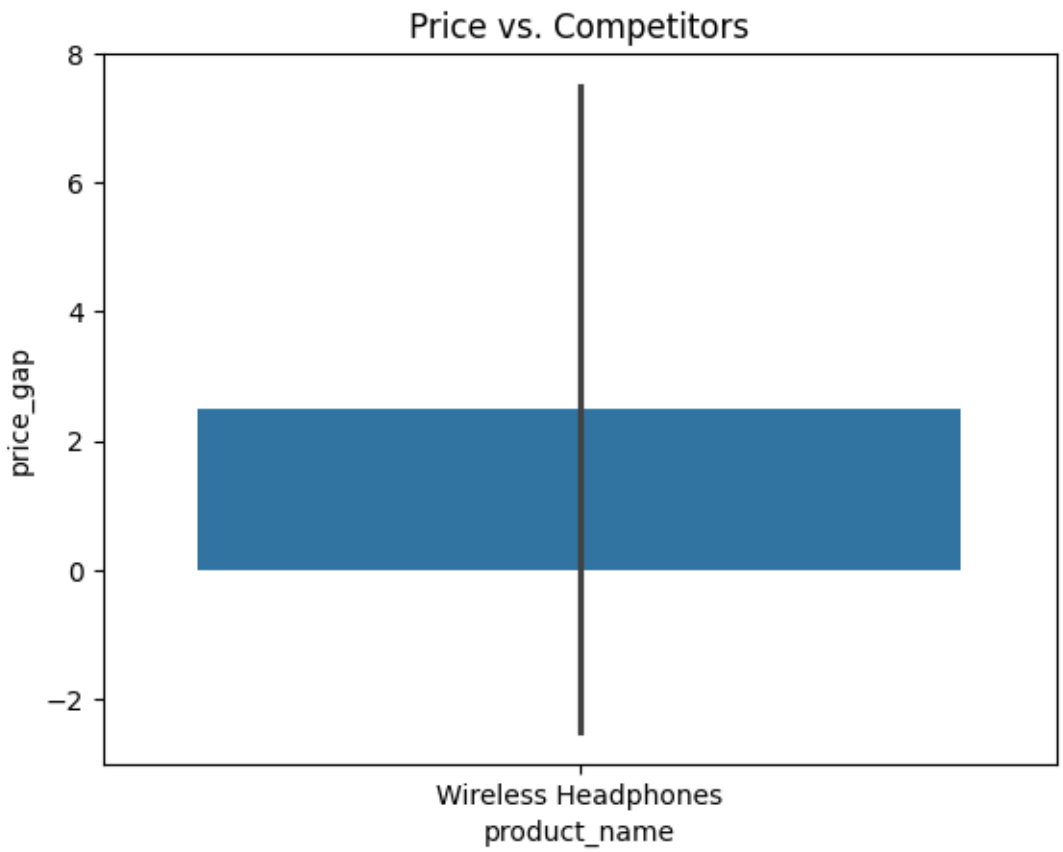
- Combine products to increase perceived value
- Example: Headphones + Case

◆ **Premium Pricing**

- Increase price for inelastic products

◆ **Competitive Pricing**

- Align pricing with competitors



-

◈ 7. Revenue Impact Analysis

Evaluated the effectiveness of each strategy:

- Current Revenue
- Projected Revenue
- Percentage Change

This helps measure **real business impact of pricing decisions**.

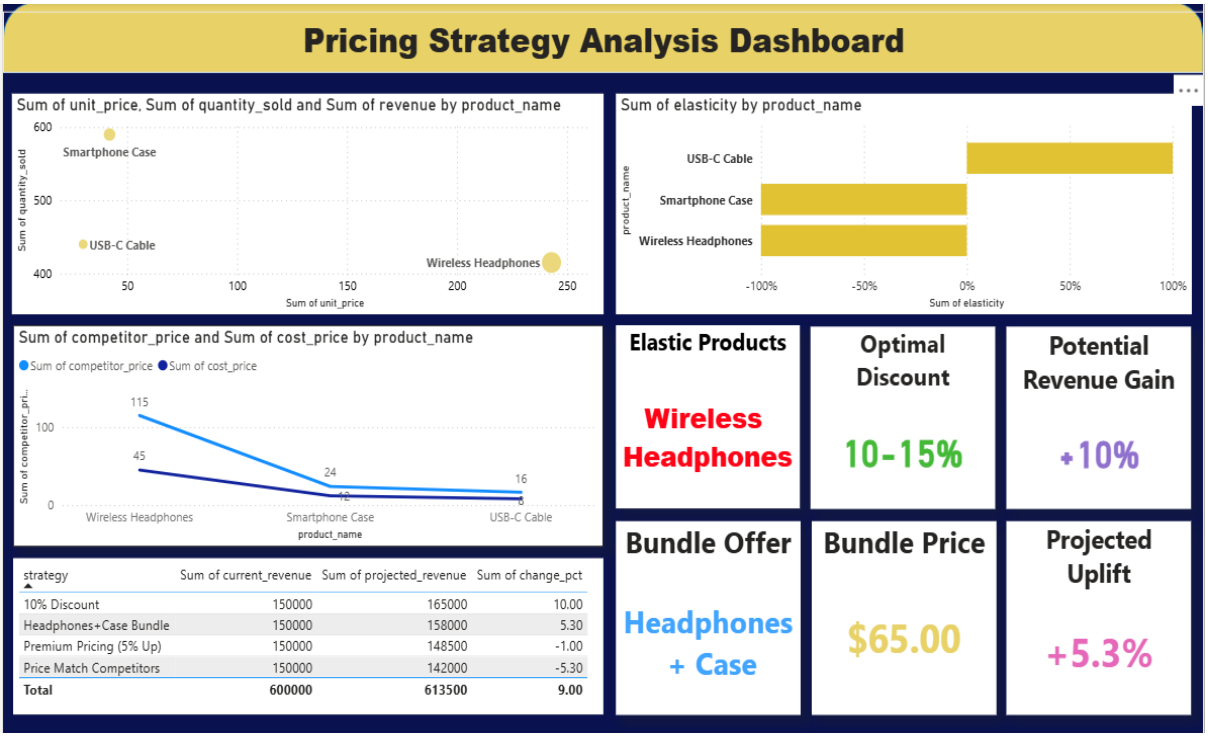
◈ 8. Dashboard Development (Power BI)

Developed an interactive dashboard with:

KPIs:

- Elastic Products
- Optimal Discount

- Potential Revenue Gain
- Bundle Offer
- Bundle Price
- Projected Uplift



strategy	Sum of current_revenue	Sum of projected_revenue	Sum of change_pct
10% Discount	150000	165000	10.00
Headphones+Case Bundle	150000	158000	5.30
Premium Pricing (5% Up)	150000	148500	-1.00
Price Match Competitors	150000	142000	-5.30
Total	**600000**	**613500**	**9.00**

Visualizations:

- Scatter Plot → Price vs Quantity vs Revenue
- Bar Chart → Elasticity Analysis
- Line Chart → Competitor vs Cost Comparison
- Table → Strategy-wise Revenue Impact

🔷 9. Insights & Business Decisions

Key findings:

- Elastic products benefit from discounts
- Bundling increases overall revenue
- Premium products generate higher margins
- Competitor pricing influences demand significantly

🔷 10. Recommendations

Based on analysis:

- Use targeted discounts for elastic products
- Promote bundling strategies
- Apply premium pricing where demand is stable
- Continuously monitor competitor pricing
- Focus on profit optimization, not just revenue

💎 FINAL WORKFLOW SUMMARY

Problem Definition → Data Collection → Data Cleaning → Feature Engineering → EDA → Strategy Design → Impact Analysis → Dashboard → Insights → Recommendations